

JOSE ANGEL SANCHEZ GOMEZ

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PROFESSIONAL EXPERIENCE

Assistant Professor <i>Department of Statistics, University of California at Riverside (UCR)</i>	August 2023 - current <i>Riverside, CA</i>
Data Science for the Social Good Fellow <i>German Research Center for Artificial Intelligence (DFKI)</i>	June-August 2022 <i>Kaiserslautern, Germany</i>

EDUCATION

University of North Carolina at Chapel Hill, USA <i>Ph.D. in Statistics</i>	August 2018 - June 2023
The University of Guanajuato, Mexico <i>B.Sc. in Mathematics</i>	August 2013 - June 2018

RESEARCH MANUSCRIPTS

Peer-Reviewed Articles

- **Sánchez Gómez, J.**, Zhang, E., Liu, Y. (2025) *Effective Permutation Tests for Differences Across Multiple High-Dimensional Correlation Matrices*. Journal of Computational and Graphical Statistics, 1–10 <https://doi.org/10.1080/10618600.2025.2550527>.
- Cordero, J. M., Henn, T., Holtel, F., **Sánchez Gómez, J.**, Arenas, D., Sipka, A. Vollmer, S. (2025). *Developing Effective and Value-Aligned AI Tools for Journalists: 12 Critical Questions to Reflect upon*. Accepted in Journalism Practice <https://doi.org/10.1080/17512786.2025.2465894>.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. *Hub Detection in Gaussian Graphical Models*. Journal of the American Statistical Association 120(552), 2397-2409. <https://doi.org/10.1080/01621459.2025.2453250>.
- Griffith, A., **Sánchez Gómez, J.**, Castillo, K. (2024). *Hurricane Irma Linked to Coral Skeletal Density Shifts on The Florida Keys Reef Tract*. Integrative And Comparative Biology. <https://doi.org/10.1093/icb/icae128>
- Chowdhury, S., Clause, N., Memoli, F., **Sánchez Gómez, J.**, Wellner, Z. (2020) *New Families of Stable Simplicial Filtration Functors*. Topology and its Applications, 279(1), 107254. <https://doi.org/10.1016/j.topol.2020.107254>

In Preparation and In Progress

- **Sánchez Gómez, J.**, O'Rourke, H., Brewer, G. (2026) *Estimating Moderator Effects on Multiple Subject Intensive Longitudinal Studies*. In preparation since October 2024.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. (2026) *Joint Eigenspace Analysis: Perturbation Theory and Applications*. In progress since January, 2026.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. (2026) *Identifying Common Hubs in Multiple Gaussian Graphical Models*. Under review at the 43rd International Conference on Machine Learning (ICML-2026).

- Cai, X., Gao, Q., **Sánchez Gómez, J.**, Au, K. (2026) *Cluster-assisted Gene Isoform Quantification: Theory and Practice*. In preparation since October 2023.
- Seongoh, P., **Sánchez Gómez, J.** (2026) *Transfer Learning for High-Dimensional Gaussian Graphical Models*. In preparation since September, 2024.
- **Sánchez Gómez, J.**, Whitney, A. (2026) *Machine Learning to Predict Local Government Deficits*. In preparation since October, 2021.

TEACHING EXPERIENCE

At the University of California at Riverside

- STAT 293: High-Dimensional Statistics (Fall 2025, Fall 2024, Fall 2023)
- STAT 107: Introduction to Statistical Computing with R (Fall 2025, Winter 2025, Winter 2024)
- STAT 155: Probability and Statistics for Science and Engineering (Winter 2024, Spring 2024)

PRESENTATIONS

- Invited Speaker “Identifying Common Hubs in Multiple Gaussian Graphical Models” at the Joint Statistical Meeting of the American Statistical Association in Nashville, Tennessee, August, 2025.
- Invited Speaker “Hub Detection in Gaussian Graphical Models” at the at Newport Beach, CA, November, 2024.
- Invited Speaker “Hub Detection in Gaussian Graphical Models” at the 7th International Conference on Econometrics and Statistics (EcoSta2024) at Beijing Normal University, July, 2024.
- Invited Speaker “Hub Detection in Gaussian Graphical Models” at the Institute of Mathematical Statistics Asia-Pacific Rim Meeting (IMS-APRM) at University of Melbourne, January, 2024.
- Presentation “Estimating hubs in Gaussian Graphical Models” in the Statistics Colloquium at the Ohio State University, February, 2023.
- Presentation “Estimating hubs in Gaussian Graphical Models” in the Statistics Colloquium at Southern Methodist University, January, 2023.
- Presentation “Estimating hubs in Gaussian Graphical Models” in the Statistics Colloquium at UC-Riverside, January, 2023.
- Presentation “Estimating hubs in Gaussian Graphical Models” in the Biomedical Research Facility at UC-San Diego, January, 2023.
- Presentation “Estimating hubs in Gaussian Graphical Models” in the Mathematics Colloquium at San Francisco State University, December, 2022.
- Presentation “Searching for New Topological Filtrations, Data” in the *Joint Meeting of the Mexican and Colombian Mathematical Societies* hosted in Barranquilla, Colombia, June, 2018.
- Poster “A First Approach to a General Theory of Filtration Functors” at the *2018 Latinx in the Mathematical Sciences Conference* at the Institute for Pure and Applied Mathematics (IPAM), Los Angeles, USA, March, 2018.

HONORS AND AWARDS

2019 Cambanis-Hoeffding-Nicholson Award

For superior academic performance in the UNC-Chapel Hill doctoral program in Statistics.